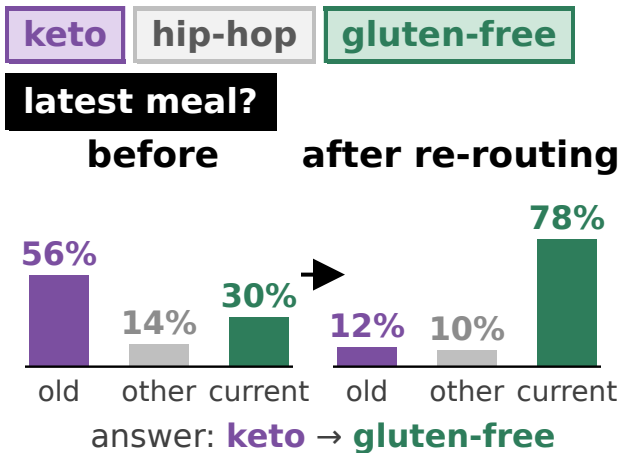


Re-routing attention from the old assignment to the current one repairs the answer, without retraining

Held-out accuracy on 960 CICM dialogues per model; heads and routing margin chosen on 240 separate calibration dialogues and frozen first.

(a) The intervention

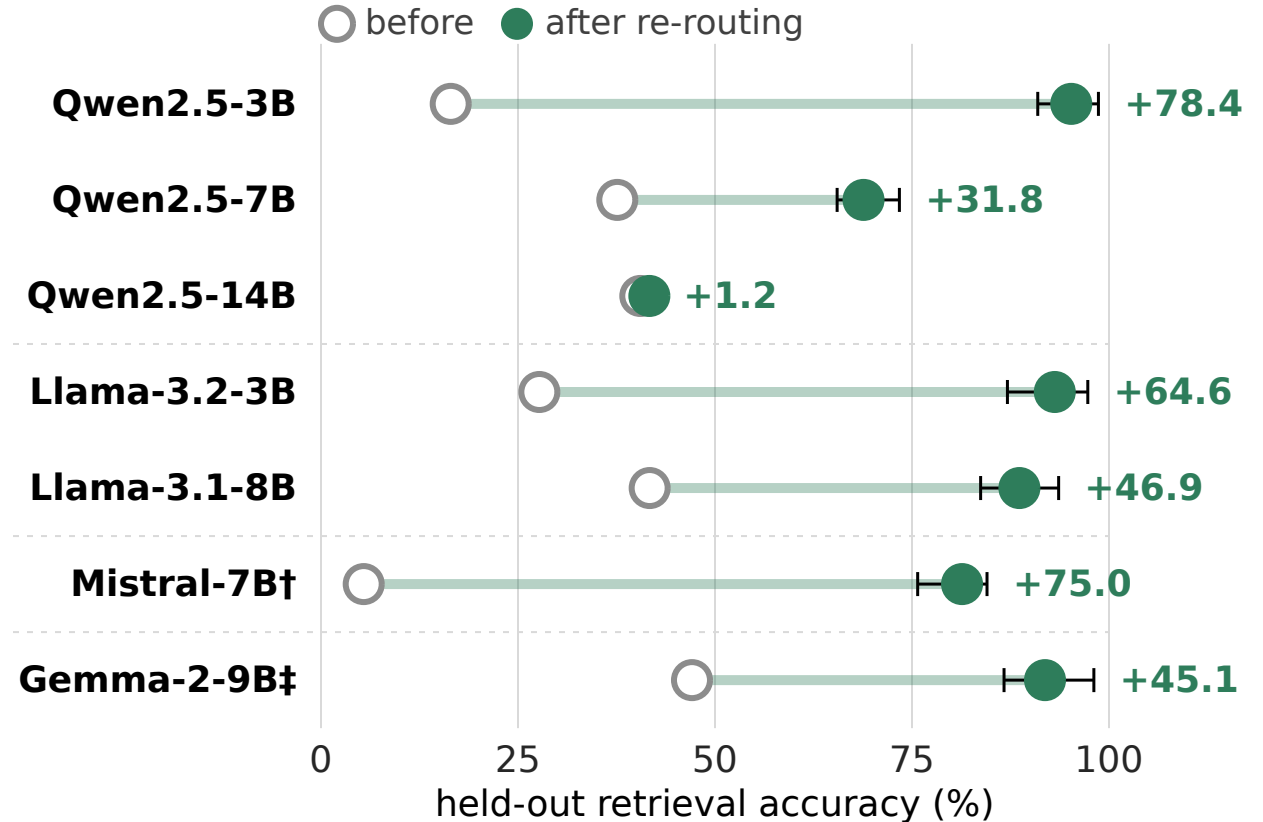
answer position, selected heads only



Lower the query's attention logits to the **old assignment**, raise them for the **current one**, in heads chosen on calibration dialogues. **Weights unchanged.** Bars are schematic; measured attention shares are in the mechanism figure.

(b) Before and after re-routing, seven models

held-out retrieval accuracy (%); label = gain in points; whisker = 95% clustered bootstrap interval of the gain



Controls (Qwen2.5-7B): the opposite route lowers accuracy by 16 points; 16 random-position routes change it by 0.2 points; 64 layer-matched random head sets fall far below the targeted set. Already-correct answers preserved: 98–100% in every model. † Mistral had only 52 baseline-correct dialogues (preregistered minimum 100), so its preservation estimate is imprecise. ‡ Gemma's no-op identity check was excluded from its gate. Qwen2.5-7B used a fixed bias on 8 heads; the other models an adaptive margin on model-specific heads found by causal screening. Source: results/stage_r/PAPER_REPAIR_SYNTHESIS.md.